

CYBER SECURITY TASK 02

PART B – REAL-WORLD CASE STUDY

NAME : BHAKTI MAHADEV PARHAD

Facebook Data Leak

What Happened?

In 2021, it was discovered that the personal information of more than 530 million Facebook users had been exposed online. The leaked data included names, phone numbers, email addresses, locations, birth dates, and other profile information from users in over 100 countries.

According to Facebook, the data was collected by attackers who exploited a feature that allowed them to scrape publicly available information before the vulnerability was fixed in 2019. Although Facebook stated that its systems were not directly hacked, the exposed data was later shared freely on hacker forums, making it accessible to anyone.

Which Part of the CIA Triad Was Affected?

The primary part of the CIA Triad that was affected was Confidentiality.

Confidentiality is about ensuring that sensitive information is accessible only to authorized users. In this case, millions of users' personal details became publicly available without their permission, violating the confidentiality of their data.

What Was the Impact?

The Facebook data leak had several serious consequences:

- Personal information of over 530 million users was exposed.
- Cybercriminals could use the leaked data for phishing attacks, identity theft, and scams.
- Users lost trust in Facebook's ability to protect their information.
- Facebook faced legal investigations, regulatory scrutiny, and damage to its reputation.

How Could It Have Been Prevented?

The incident could have been prevented by :

- Regularly identifying and fixing security vulnerabilities.
- Limiting automated data scraping by implementing stronger protections.
- Monitoring unusual activity to detect large-scale data collection.
- Applying stronger privacy controls for user information.
- Conducting regular security audits and vulnerability assessments.
- Educating users about protecting their personal information and enabling security features such as Multi-Factor Authentication (MFA).

Conclusion :

The Facebook Data Leak highlights the importance of protecting users' personal information. It demonstrates how weaknesses in system design and inadequate safeguards can expose sensitive data on a massive scale. Organizations should continuously improve their security measures to protect user privacy and maintain trust.